LUND UNIVERSITY

LTH FACULTY OF ENGINEERING

Department of Building and Environmental Technology

Report on Energy Usage and Thermal Comfort Simulation using Grasshopper.

1 Introduction

The assignment is aimed at familiarising with ClimateStudio, focusing on its inputs and outputs and how different parameters impact building energy needs and thermal comfort. This individual assignment is a simplified version of my living space; I have maintained computational effort by estimating the material properties and settings of the room.

I have a baseline simulation for the initial inputs. Then, choose three parameters and run these through three settings to test their influence on the Energy Use Intensity and thermal comfort of the building. Some possible parameters to be changed include insulation thickness, thermal mass, room orientation, room shape, and context variables such as outdoor environment.

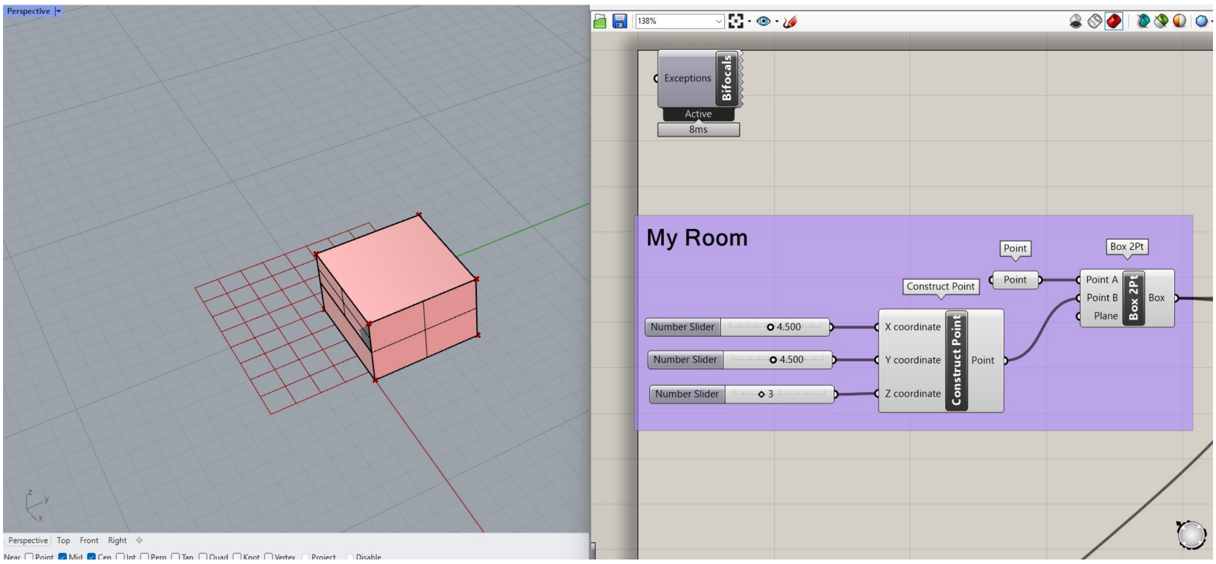
2 Methodology

Base Case:

2.1.1 1. Preparing the Base Model in Rhino

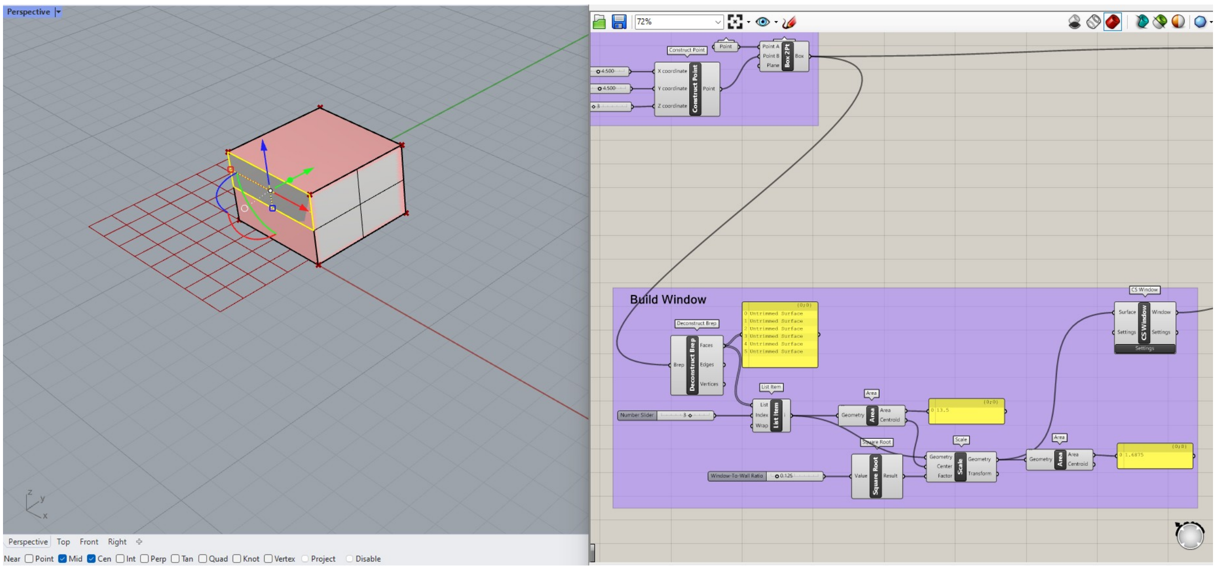
For Rhino's base model, consider a 4.5m × 4.5m room and 3m high. In Rhino, model this room directly with the Box or Solid tool either by corner points or by typing the dimensions in. Make sure the geometry is defined as a BREP - Boundary Representation - as it will be used for energy analysis in later steps.

Alternatively, create the same room parametrically in Grasshopper. Using the Construct Point component, define the corner points at (0,0,0), (4.5,0,0), (4.5,4.5,0) and (0,4.5,0) of the room. Connect the points to create the base of the room using the Brep or Box component. Set the height of the room to 3m. This will provide a parametric definition of the room dimensions.

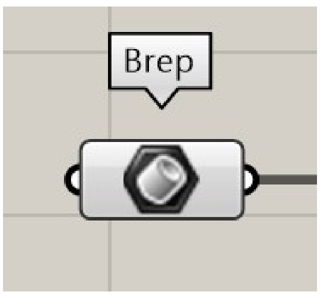


To add a window on the south side of the room in Rhino, create a surface with dimensions of 1.2m by 4m, positioned at a sill height of 1.6m. Use the **Rectangle** tool to draw the window on the south face, setting the lower-left corner at (0,1.6,1.6) and defining the width and height.

In Grasshopper, use the **Construct Point** component to define the window's corner points and the **Brep** component to create the window surface, ensuring it aligns with the room geometry at the specified sill height.



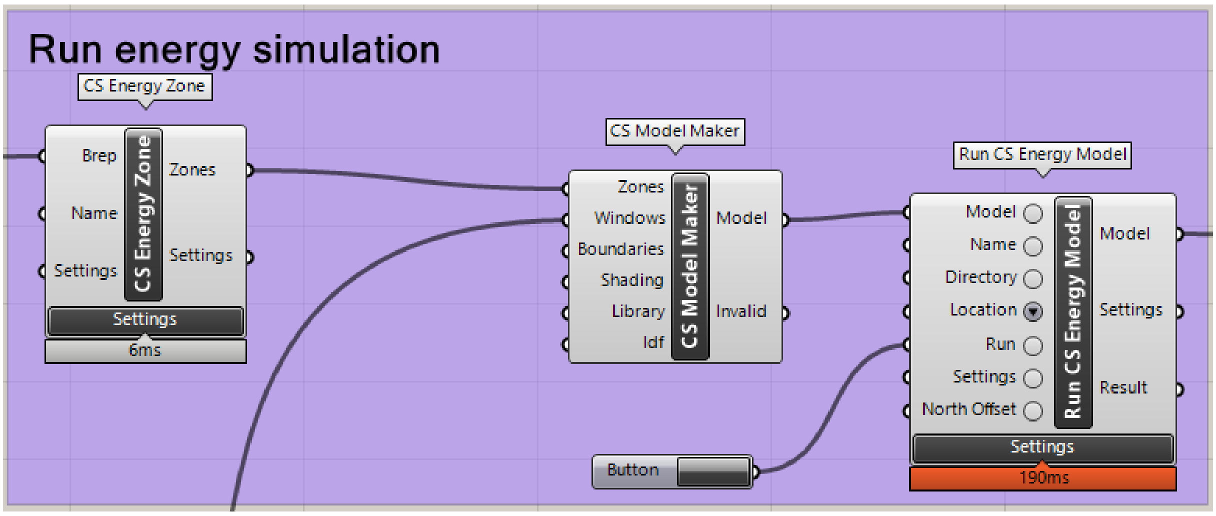
2. Setting Up Grasshopper for Parametric Control



Once the Rhino model is ready, switch to **Grasshopper** to add parametric control and data input flexibility. Grasshopper allows you to import the Rhino geometry (BREP) and modify it dynamically throughout the simulation workflow. By adding the Brep component in Grasshopper, you can reference the Rhino geometry directly. This integration between Rhino and Grasshopper allows you to use the building geometry as an input for the ClimateStudio components that will be added in the next step. (Keep the model simple, clean, and define each zone or space appropriately, as this will be important later on to be able to achieve the energy analysis)

2.1.2 3. Integrating ClimateStudio Components

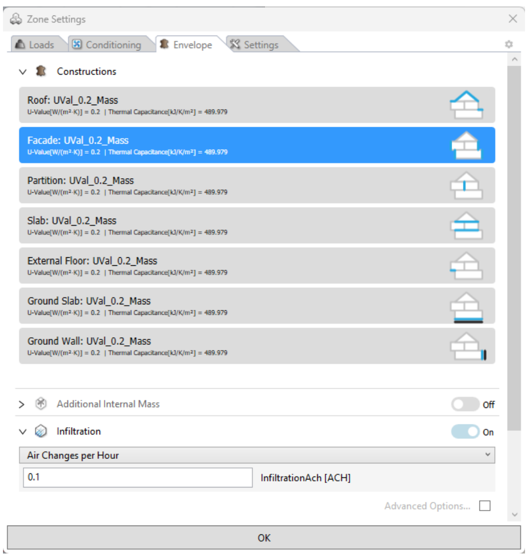
With the geometry in place, the next step is to integrate ClimateStudio components within Grasshopper. First, use the **CS Energy Zone** component to define the energy zones by connecting the BREP input. Set the base case simulation parameters by turning on **Heating Conditioning** and applying **Envelope Constraints** with a U-value of 0.2 for all. After defining the energy zone, connect it to the **CS Model Maker** component to compile the energy zones into a simulation-ready model. To run the simulation, use the **Run CS Energy Model** component, which will execute the energy calculations based on the defined inputs and parameters. Finally, add the **Load CS Energy Results** component to extract and visualise the simulation outputs, such as heating energy demand and overall energy performance.



2.1.3 Steps:

➔ **CS Energy Zone: Define the Energy Zone**

- ◆ Add the **CS Energy Zone** component to Grasshopper.
- ◆ Connect the BREP (geometry) input to the **CS Energy Zone** component.



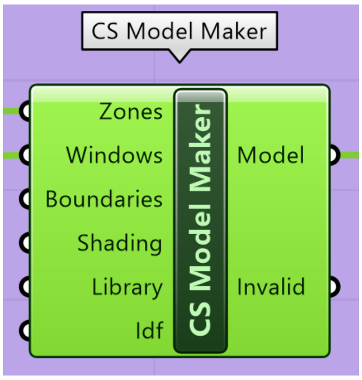
◆ Set the base case parameters:

➔ Enable **Heating Conditioning**.

- Enable **Heating Conditioning**.

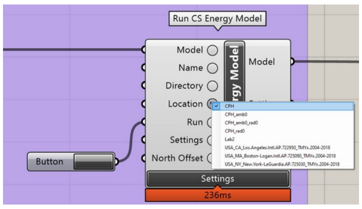
- Set **Envelope Constraints** with U-value of 0.2 for the Mass.

➔ **CS Model Maker: Create the Simulation Model**

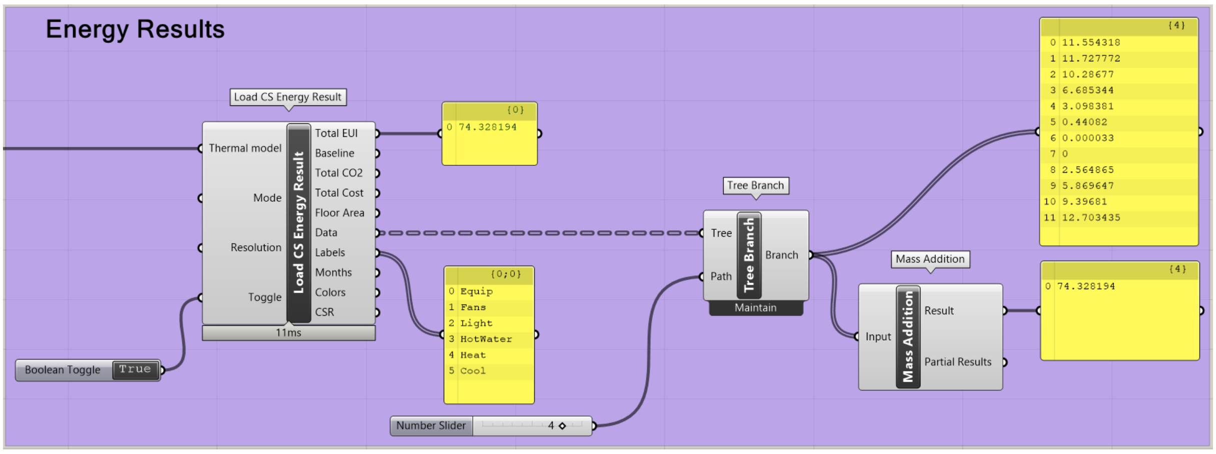


- Add the **CS Model Maker** component to Grasshopper.
- Connect the **CS Energy Zone** component output to the **CS Model Maker**.
- This will compile the energy zones into a simulation-ready model for ClimateStudio.

➔ **Run CS Energy Model: Execute the Energy Simulation**

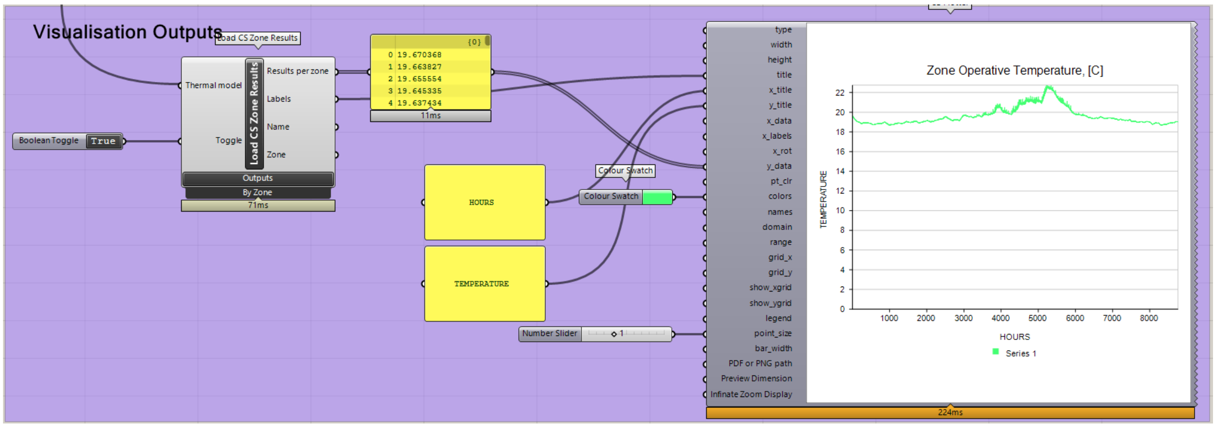


- Add the **Run CS Energy Model** component to Grasshopper.
- Connect the **CS Model Maker** to the **Run CS Energy Model**.
- Set the desired simulation parameters, such as simulation period and step size.
- In the **CS Energy Model** component, load the weather file by selecting **CPH (Copenhagen)** from the location dropdown menu to incorporate accurate climate data into the simulation.

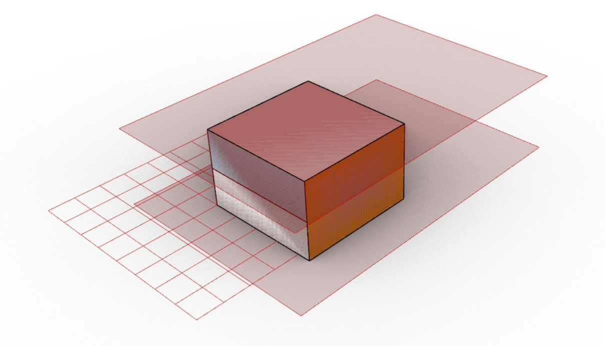


The **Load CS Energy Result** component is used to import simulation data such as total energy use. The **Tree Branch** component filters specific energy categories like heating, cooling, and lighting, while the **Mass Addition** component sums these filtered results. A **Boolean Toggle** activates the simulation, and a **Number Slider** adjusts parameters. The total energy consumption, displayed in a panel, allows for easy visualisation and analysis of specific energy usage categories from the simulation.

➔ **Visualisation Outputs**



The **Load CS Zone Results** component is used to import zone-specific results, such as zone operative temperature over time. The **Thermal Model** from ClimateStudio is connected to the **Load CS Zone Results** component, and data such as **hours** and **temperature** are filtered for visualisation. The results are plotted on a graph displaying temperature versus hours, showing the zone's thermal performance over a specific period. A **Colour Swatch** is applied to customise the graph's appearance for better visual clarity.

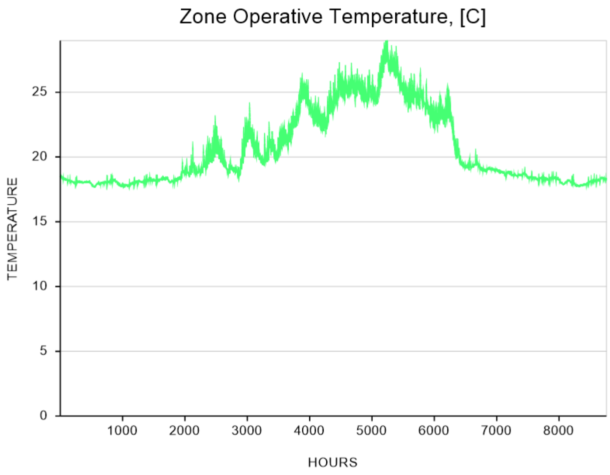


3 Results

3.: Base Case:

- **Load** with people of 0.08 [P/m2]

- **Envelope Constraints** with U-value of 0.2 for the Mass.
- **Lighting** with people of 1[W/m2]



Results:

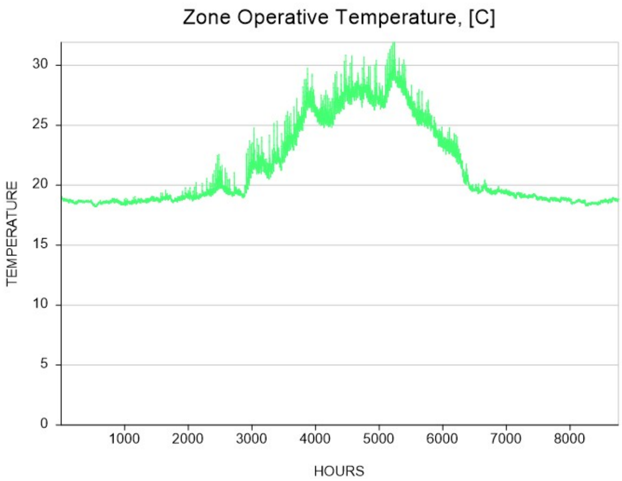
- Total Energy Use Intensity (EUI):

  45.44 kWh/m²

The **Zone Operative Temperature** graph shows substantial fluctuations throughout the simulation. Initially, the temperature remains around **20°C**, but it begins to rise after the 1000th hour, peaking between **35°C** and **40°C** around the 4000th to 6000th hour. This period indicates a high thermal load. Afterward, temperatures drop gradually and stabilise around **20°C**. The **Total Energy Use Intensity (EUI)** for the zone is **45.44 kWh/m²**, reflecting the energy performance over the simulation period.

3.2 Rotation: (90°, 180°, 270°)

3.3 Case 1 Orientation 90°:

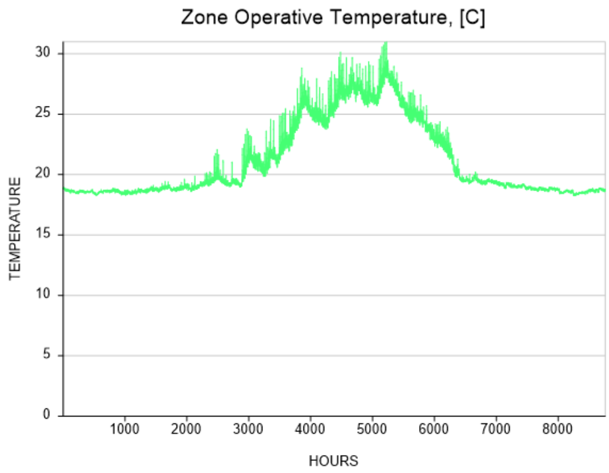


- Total Energy Use Intensity (EUI):

  82.857 kWh/m²
- **Building Orientation**: 90 degrees

3.4 The Zone Operative Temperature graph shows temperature variations over time with significant peaks around the 3000th to 6000th hour, where temperatures rise to nearly 30°C. The graph indicates fluctuations in thermal conditions, with a gradual return to approximately 20°C. Additionally, the building's Total Energy Use Intensity (EUI) is recorded as **84.34 kWh/m²**. The orientation is set at **90 degrees**, likely impacting solar exposure and heat gain, contributing to the temperature and energy performance results seen in the simulation.

3.5 Case 2 Orientation 180°:

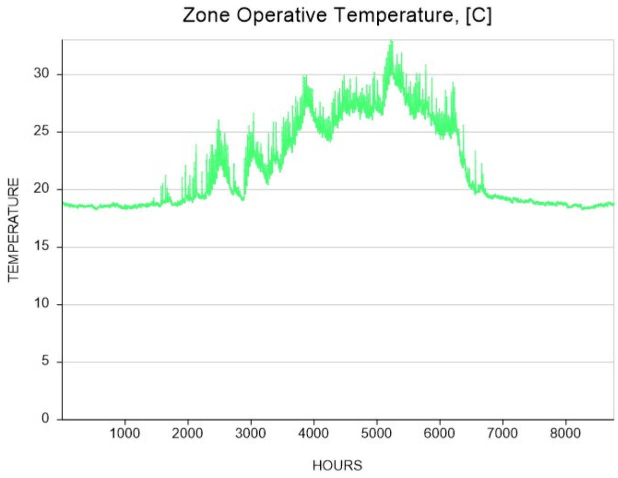


- Total Energy Use Intensity (EUI):

  88.51 kWh/m²
- **Building Orientation:** 180 degrees

The Zone Operative Temperature graph shows temperature changes over time, with a peak temperature of nearly 30°C between the 4000th and 6000th hour, followed by a gradual decline to around 20°C. These fluctuations reflect the thermal behaviour of the building over time. The building's Total Energy Use Intensity (EUI) is recorded at **88.51 kWh/m²**, indicating its energy performance. The building's orientation is set to **180 degrees**, which likely affects its exposure to solar radiation and contributes to the observed temperature and energy usage trends.

3.6 Case 3 Orientation 270°:

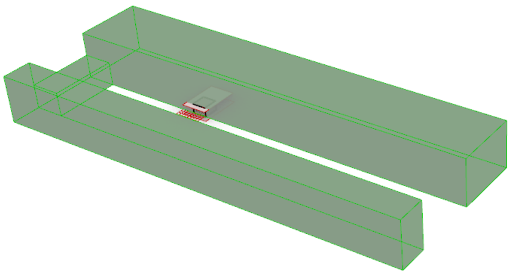


- Total Energy Use Intensity (EUI):

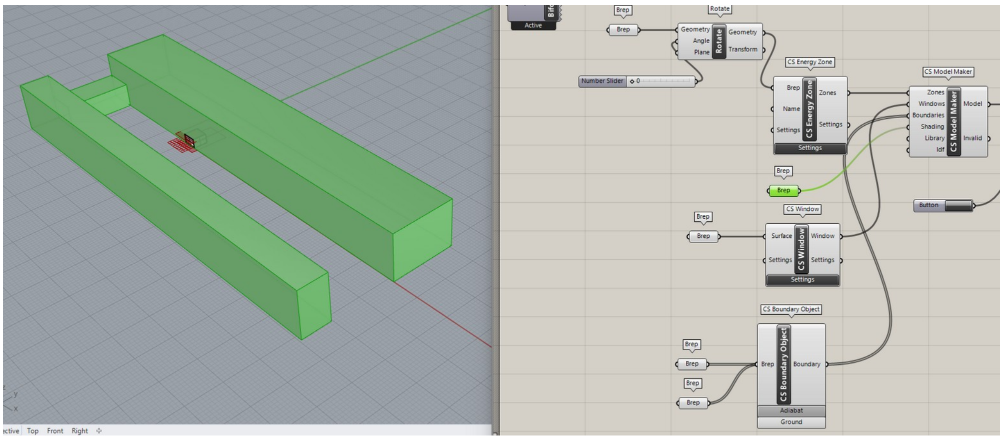
  83.24 kWh/m²
- **Building Orientation:** 270 degrees

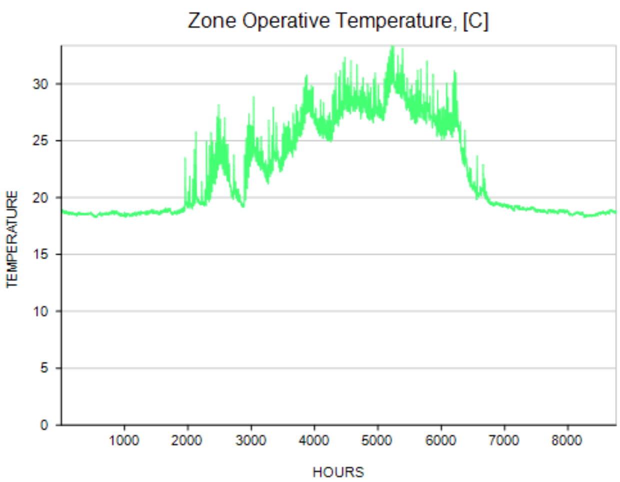
The **Zone Operative Temperature** graph shows significant variations over time, with temperatures rising to nearly **30°C** between the 3000th and 6000th hour and then gradually decreasing back to around **20°C**. These fluctuations reflect the impact of solar gain and internal loads on the building. The building's **Total Energy Use Intensity (EUI)** is recorded at **83.24 kWh/m²**, indicating its overall energy consumption. The building's orientation is set to **270 degrees**, influencing its solar exposure and affecting the observed temperature and energy performance trends.

3.7 Outdoor Environments Case: (settings: Modelling Building, etc)



The building environment, acting as shading, plays a crucial role in controlling solar gain and enhancing energy efficiency. In ClimateStudio's **CS Model Maker** environmental elements like surrounding buildings or trees can function as shading, regulating the amount of sunlight entering the building, reducing overheating, lowering cooling loads, and optimising both thermal comfort and energy performance.





- Total Energy Use Intensity (EUI):

  82.857 kWh/m²

3.8 The Zone Operative Temperature graph displays significant temperature variations over time. Initially, the temperature is stable around 20°C. Around the 2000th hour, temperatures start rising and peak between 30°C and 35°C around the 4000th to 6000th hour, indicating a period of increased thermal load. Afterward, temperatures gradually decrease and stabilise back around 20°C. Additionally, the Total Energy Use Intensity (EUI) for the zone is **82.857 kWh/m²**, reflecting the overall energy performance during the simulation period.

4 Conclusion

In conclusion, the analysis of the Zone Operative Temperature and Total Energy Use Intensity (EUI) reveals the significant impact of orientation and external environmental factors on a building's thermal performance and energy consumption. The temperature fluctuations, particularly during peak hours, highlight the importance of managing solar gain and internal heat loads. With EUI values ranging from **83.24 to 88.51 kWh/m²**, it is evident that optimising building orientation, shading, and thermal insulation can enhance both comfort and energy efficiency, leading to better overall performance.